



Parul University

Faculty of Engineering & Technology

Department of Applied Sciences and Humanities

1st Year B.Tech Programme

Mathematics – II (303191151)

Unit 3: Laplace Transform:(Lecture Notes)

Overview:

This unit is a mathematical section to establish a base for the theory of control systems. This is a tool and it is indispensable as most of linear systems dynamics are described in a mapped space that can only be understood when the main theorems of the Laplace transform are known. The module contains only the essential results, which are explained by several examples from the area of differential equations and their solutions. Some additional mathematical details can be found in the mathematical appendix module. The correspondences of the Laplace transform are given in tabular form to be simply used for the forward and back transformation. Special focus is put on the solution of differential equations using the Laplace transform and on special signals, e.g. impulse or step.

Objectives:

When you have completed this unit you should be able to:

1. Apply the Laplace transform to differential equations.
2. Solve linear differential equations.
3. Apply the main theorems of the Laplace transform.
4. Know how useful this technique is to handle dynamical systems.

Prerequisites:

Mathematics: integrals, differential equations, complex numbers, rational and analytical functions.

Application:

- Solve vibrating spring problems with or without damping/external forces using standard methods and Laplace transform methods.
- Solve electric circuit problems with or without charge/inductance/resistance/capacitance/electromotive force, using standard methods and Laplace transform methods.
- Solve systems of differential equations by Laplace transforms.
- Solve electric networks using standard methods and Laplace transform methods.

Outline

- Laplace Transform,
- Linearity of Laplace Transform,
- Laplace transform of elementary function,
- First Shifting Theorem,
- Differentiation and Integration of Transform,
- Transform of Derivatives and Integrals,
- Evaluation of integrals by Laplace transform,
- Laplace transform of periodic function,
- Inverse Transform,
- Unit Step Function,
- Second Shifting Theorem,
- Dirac's Delta Function,
- Convolution theorem,
- Partial Fraction,
- Solution of Ordinary Differential Equations.

Weightage: 25%

Teaching Hrs.: 12 Hours

Reference Book:

1. Advanced Engineering Mathematics, Erwin Kreyszig, Willey India Edition.
2. Higher Engineering Mathematics, B. S. Grewal, Khanna Publishers.
3. Integral Transforms for Engineers, Larry S. Andrews, Bhimsen K. Shivamoggi

4. Integral Transforms and their applications, Lokenath Debnath, Dambaru Bhatta, Chapman and Hall/CRC (Unit 4)

Laplace Transform:

Let $f(t)$ be a function defined for $t \geq 0$, then the Laplace Transformation of $f(t)$ is defined as

$$L\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt = F(s)$$

Provided that integral exists. s is a parameter which may be real or complex number.

Laplace transform of elementary function:

Ex: Find the Laplace transform of 1, where $s > 0$.

Sol. We know that,

$$L\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt = F(s)$$

$$\Rightarrow L\{1\} = \int_0^{\infty} 1 \cdot e^{-st} dt = \left[\frac{e^{-st}}{-s} \right]_0^{\infty} = \left[\frac{e^{-s(\infty)}}{-s} - \frac{e^{-s(0)}}{-s} \right] = \left[0 - \frac{1}{-s} \right] = \frac{1}{s} (\because e^{-\infty} = 0)$$

$$\therefore L\{1\} = \frac{1}{s}$$

Ex: Find the Laplace transform of e^{-at} , where $s > -a$.

Sol. We know that,

$$L\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt = F(s)$$

$$\Rightarrow L\{e^{-at}\} = \int_0^{\infty} e^{-at} \cdot e^{-st} dt = \left[\frac{e^{-(s+a)t}}{-(s+a)} \right]_0^{\infty} = \left[\frac{e^{-(s+a)(\infty)}}{-(s+a)} - \frac{e^{-(s+a)(0)}}{-(s+a)} \right]$$

$$= \left[0 - \frac{1}{-(s+a)} \right] (\because e^{-\infty} = 0) = \frac{1}{(s+a)}$$

$$\therefore L\{e^{-at}\} = \frac{1}{(S+a)}$$

Also

$$\therefore L\{e^{at}\} = \frac{1}{(S-a)}, s > a$$

Ex: Show that $L\{t^n\} = \frac{\Gamma_{n+1}}{s^{n+1}}$ for $n > -1$

$$= \frac{n!}{s^{n+1}} \text{ for } n \text{ is a positive Integer and } s > 0$$

Sol. We know that,

$$L\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt$$

Here $f(t) = t^n$

$$\Rightarrow L\{t^n\} = \int_0^{\infty} t^n \cdot e^{-st} dt$$

Putting $st = x \Rightarrow dt = \frac{1}{s} dx$

Also when $t = 0, x = 0$ and $t = \infty, x = \infty$

$$L\{t^n\} = \int_0^{\infty} t^n \cdot e^{-st} dt = \int_0^{\infty} \left(\frac{x}{s}\right)^n \cdot e^{-x} \frac{dx}{s} = \frac{1}{s^{n+1}} \int_0^{\infty} x^n \cdot e^{-x} dx = \frac{1}{s^{n+1}} \int_0^{\infty} x^{(n+1)-1} \cdot e^{-x} dx$$

We know that

$$\Gamma_n = \int_0^{\infty} x^{n-1} \cdot e^{-x} dx$$

Therefore,

$$L\{t^n\} = \frac{\Gamma_{n+1}}{s^{n+1}}$$

Also, if n is a positive integer, then $\Gamma_{n+1} = n!$

So,

$$L\{t^n\} = \frac{n!}{s^{n+1}}$$

Ex: Show that $L\{\sin at\} = \frac{a}{s^2+a^2}$, **where** $s > 0$.

Sol. Here $f(t) = \sin at = \frac{e^{iat} - e^{-iat}}{2i}$

We know that,

$$L\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt$$

$$\begin{aligned} L(\sin at) &= L\left\{\frac{e^{iat} - e^{-iat}}{2i}\right\} \\ &= \frac{1}{2i} [L(e^{iat}) - L(e^{-iat})] \\ &= \frac{1}{2i} \left[\frac{1}{s - ia} - \frac{1}{s + ia} \right] \\ &= \frac{1}{2i} \left[\frac{s + ia - s + ia}{(s - ia)(s + ia)} \right] \\ &= \frac{1}{2} \left[\frac{2ia}{s^2 - i^2 a^2} \right] = \frac{a}{s^2 + a^2} \\ \therefore L(\sin at) &= \frac{a}{s^2 + a^2} \end{aligned}$$

Exercise 1: Show that $L\{\cos at\} = \frac{s}{s^2+a^2}$, **where** $s > 0$. (Hint: $\cos at = \frac{e^{iat} + e^{-iat}}{2}$)

Exercise 2: Prove that $L\{\sin hat\} = \frac{a}{s^2-a^2}$, **where** $s > |a|$. (Hint: $\sin hat = \frac{e^{at} - e^{-at}}{2}$)

Exercise 3: Prove that $L\{\cos hat\} = \frac{s}{s^2-a^2}$, **where** $s > |a|$. (Hint: $\cos hat = \frac{e^{at} + e^{-at}}{2}$)

Linearity of Laplace Transform:

If $L\{f(t)\} = \bar{f}(s)$ and $\{g(t)\} = \bar{g}(s)$, then for any constants a and b $L\{af(t) + b g(t)\} = a L\{f(t)\} + bL\{g(t)\} = a\bar{f}(s) + b\bar{g}(s)$.

Proof: We know that,

$$L\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt$$

$$\begin{aligned} L\{af(t) + b g(t)\} &= \int_0^{\infty} \{af(t) + b g(t)\}e^{-st} dt = \int_0^{\infty} af(t)e^{-st} dt + \int_0^{\infty} b g(t)e^{-st} dt \\ &= a \int_0^{\infty} f(t)e^{-st} dt + b \int_0^{\infty} g(t)e^{-st} dt = a L\{f(t)\} + bL\{g(t)\} = a\bar{f}(s) + b\bar{g}(s) \end{aligned}$$

Ex: Find the Laplace transformation of $f(t) = t^2 + \sin 3t - 2e^{-t}$.

Sol: Given that $f(t) = t^2 + \sin 3t - 2e^{-t}$

By using formula and linearity property,

$$\begin{aligned} L\{f(t)\} &= L\{t^2 + \sin 3t - 2e^{-t}\} = L\{t^2\} + L\{\sin 3t\} - L\{2e^{-t}\} \\ &= \frac{2!}{s^3} + \frac{3}{s^2 + 3^2} - \frac{2}{s + 1} \end{aligned}$$

Ex: Evaluate $L\{\cos^2 t\} = L\left(\frac{1+\cos 2t}{2}\right)$

Sol: $L\{\cos^2 t\} = L\left(\frac{1+\cos 2t}{2}\right) = L\left(\frac{1}{2}\right) + \frac{1}{2}L(\cos 2t) = \frac{1}{2s} + \frac{1}{2} \frac{s}{s^2 + 4}$

Ex: Find $L\{\cos^3 2t\}$

Sol: we know that $\cos^3 \theta = \frac{1}{4}(3 \cos \theta + \cos 3\theta)$

$$L\{\cos^3 2t\} = \frac{1}{4}L(3 \cos 2t + \cos 6t) = \frac{1}{4}\left(3 \frac{s}{s^2 + 4} + \frac{s}{s^2 + 36}\right)$$

Ex: Evaluate $L\{\sin 2t \cos 2t\}$.

Sol: $L\{\sin 2t \cos 2t\} = L\left\{\frac{1}{2}(2\sin 2t \cos 2t)\right\} = \frac{1}{2}L\{\sin 4t\} = \frac{1}{2} \frac{4}{s^2+16} = \frac{2}{s^2+16}$

Ex: Find the Laplace transformation of $f(t) = \begin{cases} -1 & 0 < t \leq 4 \\ 1 & t > 4 \end{cases}$.

Sol: We have,

$$L\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt$$

Here $f(t) = \begin{cases} -1 & 0 < t \leq 4 \\ 1 & t > 4 \end{cases}$

$$\begin{aligned} L\{f(t)\} &= \int_0^{\infty} f(t)e^{-st} dt = \int_0^4 (-1)e^{-st} dt + \int_4^{\infty} (1)e^{-st} dt = -\left[\frac{e^{-st}}{-s}\right]_0^4 + \left[\frac{e^{-st}}{-s}\right]_4^{\infty} \\ &= -\left(\frac{e^{-4s}}{-s} - \frac{e^0}{-s}\right) + \left(\frac{e^{-\infty}}{-s} - \frac{e^{-4s}}{-s}\right) = 2\frac{e^{-4s}}{s} - \frac{1}{s} = \frac{2e^{-4s} - 1}{s} \end{aligned}$$

Exercise: Find the Laplace transformation $f(t) = 2\cos 5t \sin 2t$. **Ans:** $\frac{4s^2-84}{(s^2+49)(s^2+9)}$

Exercise: Evaluate $L\{f(t)\}$ where $f(t) = \begin{cases} \sin at, & 0 < t < \frac{\pi}{a} \\ 0, & t > \frac{\pi}{a} \end{cases}$. **Ans:** $\frac{a(1+e^{-\frac{s\pi}{a}})}{s^2+a^2}$

Exercise: Find $L\{\cos^2 2t\}$. **Ans:** $\frac{s^2+8}{s(s^2+16)}$

➤ **First Shifting Theorem:**

If $L\{f(t)\} = \bar{f}(s)$, then $L\{e^{at}f(t)\} = \bar{f}(s-a) = [L\{f(t)\}]_{s \rightarrow s-a}$

Proof: By the definition of Laplace transform,

$$L(e^{at}f(t)) = \int_0^{\infty} e^{-st} e^{at} f(t) dt$$

$$= \int_0^{\infty} e^{-st+at} f(t) dt$$

$$= \int_0^{\infty} e^{-(s-a)t} f(t) dt$$

$$L(e^{at} f(t)) = F(s-a)$$

$$(\because \int_0^{\infty} e^{-st} f(t) dt = F(s))$$

Some important formula of First shifting theorem:

$$L\{e^{at}.1\} = \frac{1}{(s-a)}$$

$$L\{e^{at}.t^n\} = \frac{\Gamma_{n+1}}{(s-a)^{n+1}} \left(n = \frac{p}{q}\right) = \frac{n!}{(s-a)^{n+1}} \quad (n = \text{Integer})(n > -1)$$

$$L\{e^{at}.\sin bt\} = \frac{b}{(s-a)^2 + b^2}$$

$$L\{e^{at}.\cos bt\} = \frac{(s-a)}{(s-a)^2 + b^2}$$

$$L\{e^{at}.\sinh bt\} = \frac{b}{(s-a)^2 - b^2}$$

$$L\{e^{at}.\cosh bt\} = \frac{(s-a)}{(s-a)^2 - b^2}$$

$$L\{\sinh at f(t)\} = \frac{1}{2} [\bar{f}(s-a) - \bar{f}(s+a)]$$

and

$$L\{\cosh at f(t)\} = \frac{1}{2} [\bar{f}(s-a) + \bar{f}(s+a)]$$

Ex: Evaluate $L\{e^{-3t}(\cos 4t + 3 \sin 4t)\}$

Sol: $L\{e^{-3t}(\cos 4t + 3 \sin 4t)\} = L(e^{-3t} \cos 4t) + L(e^{-3t} 3 \sin 4t)$

First, $L(e^{-3t} \cos 4t)$

$$L(\cos 4t) = \frac{s}{s^2 + 16}$$

$$L(e^{-3t} \cos 4t) = \frac{s+3}{(s+3)^2 + 16}$$

Second, $L(e^{-3t} 3 \sin 4t)$

$$L(\sin 4t) = \frac{4}{s^2 + 16}$$

$$L(e^{-3t} \cos 4t) = \frac{4}{(s+3)^2 + 16}$$

$$\therefore L\{e^{-3t}(\cos 4t + 3 \sin 4t)\} = \frac{s+3}{(s+3)^2 + 16} + \frac{3(4)}{(s+3)^2 + 16} = \frac{s+15}{(s+3)^2 + 16}$$

Ex: Find $L(e^{-3t} \sin^2 t)$

Sol:

$$L(\sin^2 t) = L\left\{\frac{(1 - \cos 2t)}{2}\right\} = \frac{1}{2s} - \frac{s}{2(s^2 + 4)}$$

$$L(e^{-3t} \sin^2 t) = \frac{1}{2} \left[\frac{1}{s+3} - \frac{s+3}{(s+3)^2 + 4} \right]$$

Exercise: Find $L\{e^{2t} \sin^2 3t\}$. Ans: $\frac{18}{(s-2)(s^2-4s+40)}$

Exercise: Evaluate $L\{(t+1)^3 e^t\}$. Ans: $\frac{s(s^2-2s+7)}{(s-1)^4}$

Exercise: Find the Laplace transformation $f(t) = (t+1)^2 e^t$

Ans: $\frac{2}{(s-1)^3} + \frac{2}{(s-1)^2} + \frac{1}{s-1}$

Exercise: Find $L\{1 + 2\sqrt{t} + 3/\sqrt{t}\}$. Ans: $\frac{1}{s} + \frac{\sqrt{\pi}}{s} + \sqrt{\frac{\pi}{s}}$

Exercise: Find $L\{\cosh at \sin at\}$. Ans: $\frac{a(s^2+2a^2)}{(s^4+4a^4)}$

Exercise: Find $L\{e^{-t} \sin^2 t\}$

Exercise: Find $L\{\sin at \sin bt\}$

➤ **Differentiation and Integration of Transform:**

If $\{f(t)\} = F(s)$, then $L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} \{\bar{f}(s)\} = (-1)^n \frac{d^n}{ds^n} \{L\{f(t)\}\}$.

Proof: Given $L\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$

Differentiating both sides w.r.t.s, we get

$$\begin{aligned}\frac{d}{ds}[F(s)] &= \frac{d}{ds} \left[\int_0^\infty e^{-st} f(t) dt \right] \\ &= \int_0^\infty \frac{\partial}{\partial s} e^{-st} f(t) dt \\ &= \int_0^\infty (-t) e^{-st} f(t) dt \\ &= - \int_0^\infty e^{-st} (t f(t)) dt\end{aligned}$$

$$L\{t f(t)\} = (-1)^n \frac{d}{ds} [F(s)]$$

Similarly, differentiating both sides, w.r.t.s(times), one can find

$$L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} [F(s)]$$

Ex: Find the Laplace Transform of $f(t) = t^2 \sin \pi t$

Sol: We know that, $L(\sin \pi t) = \frac{\pi}{s^2 + \pi^2}$

$$\begin{aligned}L(t^2 \sin \pi t) &= (-1)^2 \frac{d^2}{ds^2} [F(s)] \\ &= \frac{d^2}{ds^2} \left(\frac{\pi}{s^2 + \pi^2} \right) \\ &= \frac{d}{ds} \left(\frac{-2s\pi}{(s^2 + \pi^2)^2} \right) \\ &= -2\pi \frac{d}{ds} \left(\frac{s}{(s^2 + \pi^2)^2} \right)\end{aligned}$$

Using $\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v u' - u v'}{v^2}$

$$\begin{aligned} \frac{d}{ds} \left(\frac{s}{(s^2 + \pi^2)^2} \right) &= \frac{(s^2 + \pi^2)^2 - 2s(s^2 + \pi^2)(2s)}{(s^2 + \pi^2)^4} \\ &= \frac{(s^2 + \pi^2) [s^2 + \pi^2 - 4s^2]}{(s^2 + \pi^2)^4} \\ &= \frac{\pi^2 - 3s^2}{(s^2 + \pi^2)^3} \\ &= -2\pi \left[\frac{\pi^2 - 3s^2}{(s^2 + \pi^2)^3} \right] \\ &= \frac{(3s^2 - \pi^2)2\pi}{(s^2 + \pi^2)^3} \end{aligned}$$

Ex: Evaluate $L \{t e^{-2t} \sin t\}$

Sol: We know that $L(\sin t) = \frac{1}{s^2 + 1}$

$$\begin{aligned} L\{t \sin t\} &= (-1) \frac{d}{ds} \left(\frac{1}{s^2 + 1} \right) = \frac{1}{(s^2 + 1)^2} (2s) = \frac{2s}{(s^2 + 1)^2} \\ \therefore L\{t e^{-2t} \sin t\} &= \frac{2(s + 2)}{((s^2 + 1)^2 + 1)^2} \end{aligned}$$

Exercise: Find $L[t e^{-t} \cos t]$. Ans: $\frac{s(s+2)}{(s^2 + 2s + 2)^2}$

Exercise: Evaluate $L[t \sin \omega t]$. Ans: $\frac{2\omega s}{(s^2 + 1)^2}$

Exercise: Evaluate $L[t e^{-4t} \sin 3t]$. Ans: $\frac{6(s+4)}{(s^2 + 8s + 25)^2}$

Change of scale Property:

If $L\{f(t)\} = F(s)$ then $L\{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right) = \frac{1}{a} [L\{f(t)\}]_{s \rightarrow \frac{s}{a}}$.

Division by t Property:

If $L\{f(t)\} = F(s)$ then $L\left\{\frac{1}{t} f(t)\right\} = \int_s^\infty F(s) ds = \int_s^\infty L\{f(t)\} ds$ provided the integral exists.

Proof: We have $L\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$

Integrating both sides w.r.t. s from s to ∞ ,

$$\begin{aligned}\int_s^\infty F(s) ds &= \int_s^\infty \left[\int_0^\infty e^{-st} f(t) dt \right] ds = \int_0^\infty \left[\int_s^\infty e^{-st} ds \right] f(t) dt \\ &= \int_0^\infty \left[\frac{e^{-st}}{-t} \right]_s^\infty f(t) dt\end{aligned}$$

$$= \int_0^\infty \frac{e^{-st}}{-t} f(t) dt = L \left\{ \frac{f(t)}{t} \right\}$$

$$\therefore L \left\{ \frac{f(t)}{t} \right\} = \int_s^\infty F(s) ds$$

Ex: Evaluate $L \left\{ \frac{t - \sinh 5t}{t} \right\}$

Sol: We know that,

$$L \left\{ \frac{f(t)}{t} \right\} = \int_s^\infty F(s) ds$$

By comparison, we get,

$$f(t) = (t - \sinh 5t)$$

$$L\{f(t)\} = L(t - \sinh 5t) = \frac{1}{s^2} - \frac{5}{s^2 - 25}$$

$$L \left\{ \frac{f(t)}{t} \right\} = \int_s^\infty \left(\frac{1}{s^2} - \frac{5}{s^2 - 25} \right) ds = \left[-\frac{1}{s} - \frac{5}{10} \log \left(\frac{s-5}{s+5} \right) \right]_s^\infty$$

$$\therefore L \left\{ \frac{f(t)}{t} \right\} = \frac{1}{s} - \frac{1}{2} \log \left(\frac{s-5}{s+5} \right)$$

Ex: Find $L \left\{ \frac{1-\cos t}{t} \right\}$

Sol: We know that,

$$L \left\{ \frac{f(t)}{t} \right\} = \int_s^\infty F(s) ds$$

$$L \{f(t)\} = L \{1 - \cos t\} = \frac{1}{s} - \frac{s}{s^2 + 1}$$

$$\begin{aligned} L \left\{ \frac{f(t)}{t} \right\} &= \int_s^\infty \frac{1}{s} - \frac{s}{s^2 + 1} ds = \int_s^\infty \frac{1}{s} - \frac{1}{2} \left(\frac{2s}{s^2 + 1} \right) ds = \left[\log s - \frac{1}{2} \log(s^2 + 1) \right]_s^\infty \\ &= \left[\log \left(\frac{s}{\sqrt{s^2 + 1}} \right) \right]_s^\infty = -\log \left(\frac{s}{\sqrt{s^2 + 1}} \right) \end{aligned}$$

Exercise: Find $L \left[\frac{e^{-3t} \sin 2t}{t} \right]$. Ans: $\cot^{-1} \frac{s+3}{2}$

Exercise: Evaluate $L \left[\frac{\sin^3 t}{t} \right]$. Ans: $\frac{3}{4} \left[\cot^{-1} s - \frac{1}{3} \cot^{-1} \frac{s}{3} \right]$

Exercise: Evaluate $L \left[\frac{\cos 2t - \cos 3t}{t} \right]$. Ans: $\frac{1}{2} \log \left[\frac{s^2 + 9}{s^2 + 4} \right]$

➤ **Transform of Derivatives and Integrals:**

• **Laplace transform of the derivative:**

$$L\{f'(t)\} = s L\{f(t)\} - f(0),$$

$$L\{f''(t)\} = s^2 L\{f(t)\} - sf(0) - f'(0),$$

Similarly, in general

$$L\{f^n(t)\} = s^n L\{f(t)\} - s^{n-1} f(0) - s^{n-2} f'(0) \dots - f^{(n-1)}(0).$$

- **Laplace transform of the integral of a function:**

If $L\{f(t)\} = F(s)$, then $L\left[\int_0^t f(u)du\right] = \frac{1}{s}F(s)$

Proof: The function $f(t)$ should be integrable in such a way that

$$g(t) = \int_0^t f(u)du$$

Is of exponential order. Then $g(0) = 0$ and $g'(t) = f(t)$. Therefore,

$$L\{g'(t)\} = s L\{g(t)\} - g(0) = s L\{g(t)\}$$

and so

$$L\left[\int_0^t f(u)du\right] = L\{g(t)\} = \frac{L\{g'(t)\}}{s} = \frac{L\{f(t)\}}{s} = \frac{1}{s}F(s)$$

Ex: Find the Laplace Transform of $\int_0^t e^{-t} dt$

Sol: $L\{e^{-t}\} = \frac{1}{s+1}$

$$L\left\{\int_0^t e^{-t} dt\right\} = \frac{1}{s}L\{e^{-t}\} = \frac{1}{s(s+1)}$$

Ex: Find $L\left\{\int_0^t e^{-t} \cos t dt\right\}$

Sol: We know that $L\{\cos t\} = \frac{s}{s^2+1}$

$$\therefore L\{e^{-t} \cos t\} = \frac{(s+1)}{(s+1)^2+1} = \frac{(s+1)}{s^2+2s+2}$$

$$\therefore L\left\{\int_0^t e^{-t} \cos t dt\right\} = \frac{1}{s}\left(\frac{s+1}{s^2+2s+2}\right)$$

Exercise: Evaluate $L\left\{\int_0^t te^{-4t} \sin 3t dt\right\}$

Ans: $\frac{1}{s}\left(\frac{6(s+4)}{(s^2+8s+25)^2}\right)$

➤ **Evaluation of integrals by Laplace transform**

Ex: Evaluate $\int_0^{\infty} e^{-3t} t^5 dt$.

Sol: $\int_0^{\infty} e^{-st} t^5 dt = L\{t^5\} = \frac{5!}{s^6}$

Putting $s = 3$, we have $\int_0^{\infty} e^{-3t} t^5 dt = \frac{120}{3^6} = \frac{40}{243}$

Ex: Evaluate $\int_0^{\infty} e^{-2t} \sin^3 t dt$.

Sol: $\int_0^{\infty} e^{-st} \sin^3 t dt = L\{\sin^3 t\}$

$$= L\left\{\frac{3\sin t - \sin 3t}{4}\right\}$$

$$= \frac{3}{4} \frac{1}{s^2+1} - \frac{1}{4} \frac{3}{s^2+9}$$

$$= \frac{3}{4} \left[\frac{s^2+9-s^2-1}{(s^2+1)(s^2+9)} \right]$$

$$= \frac{6}{(s^2+1)(s^2+9)} \dots (1)$$

Putting $s = 2$ in eq (1)

$$\int_0^{\infty} e^{-2t} \sin^3 t dt = \frac{6}{(4+1)(4+9)} = \frac{6}{65}$$

Ex: Evaluate $\int_0^{\infty} t e^{-2t} \cos t dt$.

Sol: $\int_0^{\infty} t e^{-st} \cos t dt = L\{t \cos t\}$

$$= -\frac{d}{ds} L\{\cos t\}$$

$$= -\frac{d}{ds} \left(\frac{s}{s^2+1} \right)$$

$$= \frac{s^2-1}{(s^2+1)^2} \dots (1)$$

Putting $s=2$ in equation 1, we have

$$\int_0^{\infty} t e^{-2t} \cos t dt = \frac{4-1}{(4+1)^2} = \frac{3}{25}$$

Exercise: Evaluate $\int_0^{\infty} t^2 e^{-2t} \sin 3t dt$. Ans = $\frac{18}{2197}$

Show that $\int_0^{\infty} e^{-2t} \frac{\sinh t}{t} dt = \frac{1}{2} \log 3$.

➤ **Laplace Transform of periodic function:**

If $f(t)$ is a periodic function with period p , i.e. $f(t + p) = f(t)$, then

$$L(f(t)) = \frac{1}{1 - e^{-sp}} \int_0^p e^{-st} f(t) dt$$

Ex: Find L.T. of half wave rectification of $\sin \omega t$ defined by

$$f(t) = \begin{cases} \sin \omega t & \text{if } 0 < t < \frac{\pi}{\omega} \\ 0 & \text{if } \frac{\pi}{\omega} < t < \frac{2\pi}{\omega} \end{cases}$$

$$\text{Where } f\left(t + \frac{2\pi}{\omega}\right) = f(t)$$

Sol: The given function is a periodic function with period $\frac{2\pi}{\omega}$

$$L\{f(t)\} = \frac{1}{1 - e^{-\frac{2\pi s}{\omega}}} \int_0^{\frac{2\pi}{\omega}} f(t) e^{-st} dt$$

$$= \frac{1}{1 - e^{-\frac{2\pi s}{\omega}}} \left[\int_0^{\frac{\pi}{\omega}} e^{-st} \sin \omega t dt + \int_{\frac{\pi}{\omega}}^{\frac{2\pi}{\omega}} e^{-st} (0) dt \right]$$

$$= \frac{1}{1 - e^{-\frac{2\pi s}{\omega}}} \left[\frac{e^{-st}}{s^2 + \omega^2} (-s \sin \omega t - \omega \cos \omega t) \right]_0^{\pi/\omega}$$

$$= \frac{1}{1 - e^{-\frac{2\pi s}{\omega}}} \left[\frac{e^{-s\pi/\omega} \omega}{s^2 + \omega^2} + \frac{\omega}{s^2 + \omega^2} \right] = \frac{1}{1 - e^{-\frac{2\pi s}{\omega}}} \left[\frac{\omega e^{-s\pi/\omega} + \omega}{s^2 + \omega^2} \right]$$

Ex: Find the Laplace transform of the function $f(t) = |\sin \omega t|; t \geq 0$

Sol. The given function is a periodic function of period $\frac{\pi}{\omega}$

$$L\{f(t)\} = \frac{1}{1 - e^{-ps}} \int_0^p e^{-st} f(t) dt = \frac{1}{1 - e^{-\frac{\pi s}{\omega}}} \int_0^{\frac{\pi}{\omega}} e^{-st} |\sin \omega t| dt$$

$$\begin{aligned}
&= \frac{1}{1 - e^{-\frac{\pi s}{\omega}}} \left[\frac{e^{-st}}{s^2 + \omega^2} (-s \sin \omega t - \omega \cos \omega t) \right]_0^{\frac{\pi}{\omega}} \\
&= \frac{1}{1 - e^{-\frac{\pi s}{\omega}}} \left[\frac{\omega e^{-\frac{\pi s}{\omega}}}{s^2 + \omega^2} + \frac{\omega}{s^2 + \omega^2} \right]
\end{aligned}$$

Exercise: Find the Laplace transform of $f(t) = \begin{cases} k & 0 < t < a \\ -k & a < t < 2a \end{cases}$ and

$$f(t + 2a) = f(t).$$

Ans: $\frac{k}{s} \tanh\left(\frac{as}{2}\right)$

Inverse Laplace Transform:

If $L\{f(t)\} = \int_0^\infty f(t)e^{-st}dt = F(s)$ then $L^{-1}\{F(s)\} = f(t)$

Some basic Inverse Laplace Transform Formula:

$$L^{-1}\left\{\frac{1}{s}\right\} = 1$$

$$L^{-1}\left\{\frac{1}{s^2}\right\} = t \text{ and } L^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{n!}$$

$$L^{-1}\left\{\frac{1}{s-a}\right\} = e^{at} \text{ and } L^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}$$

$$L^{-1}\left\{\frac{1}{s^2 + a^2}\right\} = \frac{\sin at}{a}$$

$$L^{-1}\left\{\frac{s}{s^2 + a^2}\right\} = \cos at$$

$$L^{-1}\left\{\frac{1}{s^2 - a^2}\right\} = \frac{\sinh at}{a}$$

$$L^{-1}\left\{\frac{s}{s^2 - a^2}\right\} = \cosh at$$

$$L^{-1}\left\{\frac{1}{(s-a)^2 + b^2}\right\} = \frac{e^{at}\sin bt}{b}$$

$$L^{-1}\left\{\frac{s-a}{(s-a)^2 + b^2}\right\} = e^{at}\cos bt$$

$$L^{-1}\left\{\frac{1}{(s-a)^2 - b^2}\right\} = \frac{e^{at}\sinh bt}{b}$$

$$L^{-1}\left\{\frac{s-a}{(s-a)^2 - b^2}\right\} = e^{at}\cosh bt$$

Partial Fractions:

Mainly four types of partial fraction are use repeatedly.

Case-1: If the denominator has non – repeated linear factors $(s-a), (s-b), (s-c)$ then

$$\text{Partial fraction} = \frac{A}{(s-a)} + \frac{B}{(s-b)} + \frac{C}{(s-c)}.$$

Case-2: If the denominator has repeated linear factors $(s-a)(n \text{ times})$, then

$$\text{Partial fraction} = \frac{A_1}{(s-a)} + \frac{A_2}{(s-a)^2} + \frac{A_3}{(s-a)^3} + \dots + \frac{A_n}{(s-a)^n}.$$

Case-3: If the denominator has non-repeated quadratic factors

$(s^2 + as + b), (s^2 + cs + d)$ then

$$\text{Partial fraction} = \frac{As+B}{(s^2+as+b)} + \frac{Cs+D}{(s^2+cs+d)}.$$

Case-4: If the denominator has repeated quadratic factors $(s^2 + as + b)(n \text{ times})$ then

$$\text{Partial fraction} = \frac{As+B}{(s^2+as+b)} + \frac{Cs+D}{(s^2+as+b)^2} + \dots (n \text{ times}).$$

$$\text{Ex: } L^{-1}\left(\frac{1}{s-7}\right) = e^{7t}$$

$$\text{Ex: } L^{-1}\left(\frac{4}{s^2-121}\right) = \frac{4}{11} \sinh 11t$$

$$\text{Ex: Find } L^{-1}\left\{\frac{6s-8}{s^2-s-6}\right\}$$

Sol: we have

$$\frac{6s - 8}{s^2 - s - 6} = \frac{6s - 8}{(s + 2)(s - 3)} = \frac{A}{(s + 2)} + \frac{B}{(s - 3)} = \frac{A(s - 3) + B(s + 2)}{(s + 2)(s - 3)}$$

$$\therefore 6s - 8 = A(s - 3) + B(s + 2) \dots (1)$$

Put $s = 3$ in above equation, we can find

$$18 - 8 = A(0) + B(5)$$

$$10 = 5B$$

$$\therefore B = 2$$

Put $s = -2$ in equation (1), one can get

$$6(-2) - 8 = A(-2 - 3) + B(0)$$

$$-12 - 8 = A(-5)$$

$$-20 = -5A$$

$$\therefore A = 4$$

By using Partial Fraction,

$$L^{-1} \left\{ \frac{6s - 8}{s^2 - s - 6} \right\} = L^{-1} \left(\frac{4}{(s + 2)} + \frac{2}{(s + 3)} \right) = 4e^{-2t} + 2e^{-3t}$$

Ex: Evaluate $L^{-1} \left\{ \log \frac{1}{s} \right\}$

Sol: Here

$$F(s) = \log \frac{1}{s} = \log 1 - \log s = -\log s$$

Differentiating with respect to s

$$-\frac{d}{ds} F(s) = -\frac{d}{ds} (-\log s) = \frac{1}{s}$$

We know that,

$$L^{-1} \left\{ -\frac{d}{ds} F(s) \right\} = t f(t)$$

$$\Rightarrow 1 = L^{-1}\left\{\frac{1}{s}\right\} = t f(t)$$

$$\Rightarrow f(t) = \frac{1}{t}$$

Exercise: Show that $L^{-1}\left(\frac{s}{s^2+4}\right) = \cos 2t$

Exercise: Prove that $L^{-1}\left(\frac{3}{s^2+6s+18}\right) = e^{-3t} \sin 3t$

Exercise: Show that $L^{-1}\left(\frac{5s+3}{(s^2+2s+5)(s-1)}\right) = -e^{-t} \cos 2t + \frac{3}{2}e^{-t} \sin 2t + e^t$

Exercise: Prove that $L^{-1}\left\{\log\left(\frac{s+1}{s}\right)\right\} = \frac{1-e^{-t}}{t}$

Exercise: Show that $L^{-1}\left\{\int_s^\infty \cot^{-1}(s+1) ds\right\} = \frac{-e^t \sin t}{t^2}$

Definition of convolution:

The $(f * g)$ is known as convolution and it is defined as

$$(f * g)(t) = \int_0^t f(u) g(t-u) du . \text{ Also } f * g = g * f$$

Convolution Theorem: If $L^{-1}\{F(s)\} = f(t)$ and $L^{-1}\{G(s)\} = g(t)$ then

$$L^{-1}\{F(s).G(s)\} = L^{-1}\{F(s)\} * L^{-1}\{G(s)\} = f(t) * g(t) = \int_0^t f(u) g(t-u) du$$

Ex: Find $1 * 1$

Sol: Let $f(t) = 1$ and $g(t) = 1$

Then by definition of convolution,

$$f(t) * g(t) = \int_0^t f(u)g(t-u)du = \int_0^t 1 \cdot 1 du = \int_0^t 1 du = [u]_1^t = t$$

$$\therefore 1 * 1 = t$$

Ex: Using convolution theorem, evaluate $L^{-1}\left\{\frac{1}{s(s^2+4)}\right\}$

Sol: $f(s) = \frac{1}{s}$ and $g(s) = \frac{1}{(s^2+4)}$

We know that

$$g(t) = L^{-1}\left\{\frac{1}{s}\right\} = 1$$

and

$$f(t) = L^{-1}\left\{\frac{1}{(s^2+4)}\right\} = \frac{1}{2} \sin 2t$$

Therefore, we get

$$g(t) = 1 \quad ; \quad f(t) = \frac{1}{2} \sin 2t$$

By convolution theorem,

$$\begin{aligned} L^{-1}\left\{\frac{1}{s(s^2+4)}\right\} &= \int_0^t f(u)g(t-u)du = \int_0^t \frac{1}{2} \sin 2u \cdot (1)du = \frac{1}{2} \left[-\frac{\cos 2u}{2} \right]_0^t \\ &= \frac{1}{4} [-\cos 2t + \cos 0] = \frac{1}{4} [1 - \cos 2t] \end{aligned}$$

Exercise: Find $\cos \omega t * \sin \omega t$. Ans: $\frac{t \sin \omega t}{2}$

Exercise: Apply Convolution Theorem to find $L^{-1}\left\{\frac{s^2}{s^4-a^4}\right\}$. Ans: $\frac{\sin at}{2a} + \frac{e^{at}-e^{-at}}{2a}$

Exercise: Find the convolution of e^t and e^{-t} . Ans: $\sin ht$

➤ **Unit Step Function(or Heaviside's unit function):**

Find the Laplace transform of unit step function

Or

Show that $L\{u(t-a)\} = \frac{e^{-as}}{s}$ **where** $u(t-a) = \begin{cases} 0, & t < a \\ 1, & t \geq a \end{cases}$ (This function is known as unit step function).

Proof:

$$L(u(t-a)) = \int_0^{\infty} e^{-st} u(t-a) dt = \int_0^a e^{-st}(0) dt + \int_a^{\infty} e^{-st}(1) dt = \left[\frac{e^{-st}}{-s} \right]_a^{\infty}$$

$$= \frac{1}{-s} [e^{-\infty} - e^{-as}] = \frac{e^{-as}}{s}$$

$$\therefore L(u(t-a)) = \frac{e^{-as}}{s}$$

Dirac's Delta Function:

Definition: Consider the *Dirac Delta Function* f_{ε} (where $\varepsilon > 0$) which is defined by,

$$f_{\varepsilon}(t) = \delta(t-\varepsilon) = \begin{cases} \frac{1}{\varepsilon}, & a \leq t \leq a + \varepsilon \\ 0, & t > \varepsilon \end{cases}$$

Exercise: Find the Laplace transform of Dirac – delta function $\delta(t-a)$

Ans: $L\{\delta(t-a)\} = e^{-as}$

Second Shifting Theorem:

If $L\{f(t)\} = F(s)$, then $L\{f(t-a) \cdot u(t-a)\} = e^{-as}F(s) = e^{-as}L\{f(t)\}$

Proof: We know that $u(t-a) = \begin{cases} 0 & \text{for } t < a \\ 1 & t \geq a \end{cases}$

$$L\{f(t-a)u(t-a)\}$$

$$= \int_0^{\infty} e^{st} f(t-a)u(t-a) dt$$

$$= \int_0^a e^{st}(0) dt + \int_a^{\infty} e^{-st} f(t-a) dt$$

$$= \int_a^{\infty} e^{-st} f(t-a) dt$$

$$\text{Let } u = t - a \Rightarrow du = dt$$

Also if $t \rightarrow a$ then $u \rightarrow 0$ and $t \rightarrow \infty$ then $u \rightarrow \infty$

$$\int_a^{\infty} e^{-st} f(t-a) dt = \int_0^{\infty} e^{-s(u+a)} f(u) du = \int_0^{\infty} e^{-su-sa} f(u) du$$

$$= e^{-sa} \int_0^{\infty} e^{-su} f(u) du = e^{-as} F(s)$$

$$\therefore L\{f(t-a)u(t-a)\} = e^{-as} F(s)$$

Note: If a function $f(t)$ is defined as follows

$$f(t) = \begin{cases} f_1(t); & 0 < t < a \\ f_2(t); & a < t < b \\ f_3(t); & t > b \end{cases} \text{ Then } f(t) \text{ can be written as,}$$

$$f(t) = [f_1(t)u(t) - f_1(t)u(t-a)] + [f_2(t)u(t-a) - f_2(t)u(t-b)] + f_3(t)u(t-b)$$

$$\textbf{Corollary 1:} L\{f(t)u(t-a)\} = e^{-as} L\{f(t+a)\}$$

$$\textbf{Corollary 2:} L\{u(t-a)\} = L\{1 \cdot u(t-a)\} = \frac{e^{-as}}{s}$$

$$\textbf{Corollary 3:} L\{u(t-a) - u(t-b)\} = \frac{e^{-as} - e^{-bs}}{s}$$

Corollary 4:

$$L\{f(t)[u(t-a) - u(t-b)]\} = [e^{-as} L\{f(t+a)\} - e^{-bs} L\{f(t+b)\}]$$

Ex: Evaluate $L\{(t-3)^2 u(t-3)\}$

Sol. Here $f(t-a) = (t-3)^2$

$$f(t+a) = (t)^2$$

$$\therefore L\{(t-3)^2 u(t-3)\} = e^{-3s} L\{f(t+a)\} = e^{-3s} L\{(t)^2\} = \frac{2e^{-3s}}{s^3}$$

Ex: Find $L\{(t)^2 u(t-3)\}$

Sol. Here, $f(t) = t^2$

$$L\{f(t)u(t-a)\} = e^{-as} L\{f(t+a)\}$$

$$\begin{aligned} L\{(t)^2 u(t-3)\} &= e^{-3s} L\{f(t+3)\} = e^{-3s} L\{(t+3)^2\} = e^{-3s} L\{t^2 + 6t + 9\} \\ &= e^{-3s} [L(t^2) + 6L(t) + 9L(1)] \end{aligned}$$

$$\therefore L\{(t)^2 u(t-3)\} = e^{-3s} \left[\frac{2}{s^3} + \frac{6}{s^2} + \frac{9}{s} \right]$$

Ex: Evaluate $L\{e^t u(t-2)\}$

$$\text{Sol: } L\{e^t u(t-2)\} = e^{-2s} L\{e^{t+2}\} = e^{-2s} e^2 L\{e^t\} = \frac{e^{-2(s-1)}}{s-1}$$

$$\text{Exercise: Find } L\{e^{-3t} u(t-2)\}. \text{ Ans: } \frac{e^{-2s-6}}{s+3}$$

$$\text{Exercise: Evaluate } L\left\{\sin t u\left(t - \frac{\pi}{2}\right)\right\}. \text{ Ans: } e^{-\frac{\pi s}{2}} \frac{s}{s^2+1}$$

Inverse Laplace by using second shifting theorem:

$$L^{-1}\{e^{-as} F(s)\} = f(t-a)u(t-a)$$

Ex: Find $L^{-1}\left\{\frac{2e^{-s}}{s^3}\right\}$

$$\text{Sol: Here } L^{-1}\left\{\frac{2e^{-s}}{s^3}\right\} = L^{-1}\left\{e^{-s} \frac{2}{s^3}\right\}$$

By comparison, we get $a = 1$

We know that,

$$L^{-1}\left\{\frac{2}{s^3}\right\} = 2 \frac{t^2}{2!} = t^2$$

$$\therefore L^{-1}\left\{e^{-s} \frac{2}{s^3}\right\} = (t-1)^2 u(t-1)$$

Application of Laplace Transform:

Solution of Ordinary Differential Equations:

$$L\{f'(t)\} = s L\{f(t)\} - f(0),$$

$$L\{f''(t)\} = s^2 L\{f(t)\} - sf(0) - f'(0),$$

Similarly, in general Laplace transform of the n^{th} derivatives of $f(t)$

$$L(f^n(t)) = s^n L(f(t)) - s^{n-1}f(0) - s^{n-2}f'(0) \dots - f^{n-1}(0)$$

Ex: Solve the differential equation

$$y'' + 6y = 1, \quad y(0) = 2, \quad y'(0) = 0$$

Sol: Here $y'' + 6y = 1$

Taking Laplace Transform on both sides, we get

$$L(y'' + 6y) = L(1)$$

$$L(y'') + 6L(y) = \frac{1}{s}$$

$$s^2 L(y(t)) - sy(0) + 6L(y(t)) = \frac{1}{s}$$

Here, given conditions are $y(0) = 2, y'(0) = 0$

$$s^2 L(y(t)) - 2s + 6L(y(t)) = \frac{1}{s}$$

$$(s^2 + 6)L(y(t)) = \frac{1}{s} + 2s = \frac{1 + 2s^2}{s}$$

$$L(y(t)) = \frac{1 + 2s^2}{s(s^2 + 6)}$$

Apply Inverse Laplace Transformation on both sides, we get

$$y(t) = L^{-1} \left(\frac{1 + 2s^2}{s(s^2 + 6)} \right)$$

By using Partial fraction,

$$\frac{1 + 2s^2}{s(s^2 + 6)} = \frac{A}{s} + \frac{Bs + c}{(s^2 + 6)} \Rightarrow 1 + 2s^2 = A(s^2 + 6) + (Bs + c)s$$

$$\Rightarrow 1 + 2s^2 = A(s^2 + 6) + Bs^2 + cs$$

$$\text{Put } s = 0, 2(0) + 1 = A(0 + 6) \Rightarrow 1 = 6A \Rightarrow A = \frac{1}{6}$$

$$\text{Substitute } s = 1, 2(1) + 1 = A(1 + 6) + B + C \Rightarrow 3 = 7 \left(\frac{1}{6} \right) + B + C \Rightarrow 3 = \frac{7+6B+6C}{6}$$

$$\Rightarrow 18 = 7 + 6B + 6C$$

$$\Rightarrow 6B + 6C = 11 \quad (1)$$

$$\text{Put } S = -1, 2(1) + 1 = A(1 + 6) + B - C \Rightarrow 3 = 7A + B - C \Rightarrow 3 = 7 \left(\frac{1}{6} \right) + B - C$$

$$\Rightarrow 3 = \frac{7 + 6B - 6C}{6} \Rightarrow 18 = 7 + 6B - 6C$$

$$\Rightarrow 6B - 6C = 11 \quad (2)$$

Solving equation (1) and (2), we get $B = \frac{11}{6}$ and $C = 0$

$$\therefore L^{-1} \left\{ \frac{1 + 2s^2}{s(s^2 + 6)} \right\} = L^{-1} \left\{ \frac{1}{6} \left(\frac{1}{s} \right) + \frac{11}{6} \left(\frac{s}{s^2 + 6} \right) \right\} = \frac{1}{6} + \frac{11}{6} \cos \sqrt{6} t$$

Therefore, the required solution of given differential equation is,

$$y(t) = \frac{1}{6} + \frac{11}{6} \cos \sqrt{6} t$$

Ex: Solve the differential equation

$$y'' + 2y' + 5y = e^{-t} \sin t, \quad y(0) = 0, y'(0) = 1$$

Sol: Apply Laplace Transformation on both sides, we get

$$L(y'' + 2y' + 5y) = L(e^{-t} \sin t)$$

$$L(y'') + 2L(y') + 5L(y) = \frac{1}{(s + 1)^2 + 1}$$

$$s^2 L(y(t)) - sy(0) - y'(0) + 2sL(y(t)) - 2s y(0) + 5 L(y(t)) = \frac{1}{s^2 + 2s + 2}$$

Now substitute boundary conditions Immediately before solving in above equation, we find

$$s^2 L(y(t)) - s(0) - 1 + 2sL(y(t)) - 2s(0) + 5 L(y(t)) = \frac{1}{s^2 + 2s + 2}$$

$$(s^2 + 2s + 5)L(y(t)) - 1 = \frac{1}{s^2 + 2s + 2}$$

$$(s^2 + 2s + 5)L(y(t)) = \frac{1}{s^2 + 2s + 2} + 1 = \frac{1 + s^2 + 2s + 2}{s^2 + 2s + 2}$$

$$L(y(t)) = \frac{s^2 + 2s + 3}{(s^2 + 2s + 2)(s^2 + 2s + 5)}$$

Taking Laplace Transform on both sides, we get

$$y(t) = L^{-1} \left\{ \frac{s^2 + 2s + 3}{(s^2 + 2s + 2)(s^2 + 2s + 5)} \right\} = L^{-1} \left\{ \frac{2}{3(s^2 + 2s + 5)} + \frac{1}{3(s^2 + 2s + 2)} \right\}$$

$$\begin{aligned} \therefore y(t) &= L^{-1} \left\{ \frac{s^2 + 2s + 3}{(s^2 + 2s + 2)(s^2 + 2s + 5)} \right\} \\ &= L^{-1} \left\{ \frac{2}{3} \left(\frac{1}{(s+1)^2 + 4} \right) + \frac{1}{3} \left(\frac{1}{(s+1)^2 + 1} \right) \right\} \end{aligned}$$

Therefore, the required solution of given differential equation is

$$y(t) = \frac{e^{-t}}{3} [\sin 2t + \sin t]$$

Exercise: Using Laplace transformation solve the initial value Problem

$$y'' + y = \sin 2t; \quad y(0) = 2; \quad y'(0) = 1$$

$$\text{Ans: } y(t) = \frac{5}{3} \sin t - \frac{1}{3} \sin 2t + 2 \cos t$$

Exercise: Using the method of Laplace transform, solve the IVP

$$x'' + 2x' + x = e^{-t}, \quad x(0) = -1, \quad x'(0) = 1$$

Ans: $x(t) = e^{-t} \left(\frac{t^2}{2} - 1 \right)$

Exercise: Using the method of Laplace transform, solve the differential equation

$$y'' + y' - 6y = 1; \quad y(0) = 0; \quad y'(0) = 1$$

Ans: $y(t) = -\frac{1}{6} + \frac{3}{10}e^{2t} - \frac{2}{15}e^{-3t}$

Exercise: Solve $y'' + 5y' + 4y = 3\delta(t - 2)$, $y(0) = 2$, $y'(0) = -2$

Ans: $y(t) = e^{-(t-2)} - e^{-4(t-2)} u(t - 2) + 2e^t$